

VibeBuilder - Drag and Drop

Overview

The **VibeBuilder** is a professional-grade, multi-user website builder platform. Your task is to build a SaaS application that allows users to sign up and create **complete, multi-page websites** without writing code. The challenge lies in transitioning from static "forms" to a dynamic, visual editing experience, using **Selise Blocks** as the high-performance engine powering the backend.

Functional Requirements

1. Tenant & User Management (The Infrastructure)

- **User Accounts:** Every student must use the **Selise IAM Block** to handle user registration.
- **Workspace Isolation:** When a user logs in, they should only see and edit *their* websites.
- **Multi-Page Support:** Users must be able to create, name, and delete multiple pages (e.g., Home, About, Services) within a single website project.

2. The Drag-and-Drop Editor (The Core Engine)

The hallmark of a top-tier submission is the **visual builder interface**.

- **Component Library:** Provide a set of "Vibe Components" (e.g., Hero Sections, Text Blocks, Image Galleries, Contact Forms).
- **Drag-and-Drop Logic:** Users should be able to drag components into a canvas, reorder them, and edit their properties (text, colors, images) in real-time.
- **State Management:** The entire layout of a page must be serialized (converted to JSON) and saved.

3. Content & Asset Persistence

- **Dynamic Storage:** You must use appropriate Selise Blocks services to store the website structures. Your schema must be flexible enough to save complex JSON objects representing the "layout" of each page.
- **Media Handling:** All images used in the builder must be uploaded to and served from the **Selise Media Block**.
- **Real-time Saving:** Changes made in the drag-and-drop editor should be persisted to Selise Blocks so that work is never lost.

4. The Live Site Renderer

- **The "Publish" Feature:** The application must be able to render a "Live" version of the site that is separate from the "Editor" view. Live websites should be publicly available.
- **Routing:** The renderer must correctly handle navigation between the different pages created by the user (e.g., `vibe.site/user123/home` vs `vibe.site/user123/about`).

Technical Constraints

- **No Traditional Databases:** You are not allowed to use local SQL or Firebase. You must demonstrate mastery of **Selise Block APIs** to manage all data.
- **Architecture over Code:** Focus on how you structure your JSON data in the **Content Block**. A well-designed schema will allow for a much smoother drag-and-drop experience.
- **Performance:** The builder should feel snappy. Use optimized API calls to ensure that dragging an element doesn't cause significant lag.